MASTERS OF BUSINESS ADMINISTRATION

SEMESTER 1



DMBA115

MANAGERIAL ECONOMICS

Unit - 5

Revenue Analysis and Pricing Policies

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1. INTRODUCTION

In this unit, we will study revenue analysis and pricing policies. The awareness of both revenue and cost concepts are important to a managerial economist. Revenue means the sales receipts of the output produced by the firm. It depends on the market price. The amount of money which the firm receives through the sale of its output in the market is known as its revenue.

1.1 Objectives



After studying this unit, you should be able to:

- Explain the concepts of revenue
- Differentiation between different types of revenues
- Identify critically the relationship between total revenue and price elasticity of demand
- Elaborate different types of pricing practices and methods
- Evaluate various guidelines for successful pricing policy
- Analyse and judge pricing policies' impact on socio-economic conditions

2. REVENUE: MEANING AND TYPES

In this section, we will discuss the meaning of revenue along with its types. Revenue is the income received by the firm. There are three concepts of revenue – total revenue, average revenue and marginal revenue.

Total Revenue (TR)

Total revenue refers to the total amount of money that the firm receives from the sale of its products, i.e., gross revenue.

In other words, it is the total sales receipts earned from the sale of its total output produced over a given period of time. We may show total revenue as a function of the total quantity sold at a given price. Figure 7.1 depicts the graph for total revenue.

TR = f (Q). It implies that the higher the sales output, larger would be the TR.

Thus, TR = P X Q. For e.g., a firm sells 5000 units of a commodity at the rate of Rs. 5 per unit, then TR would be

TR = P x Q = 5 x 5000 = 25,000

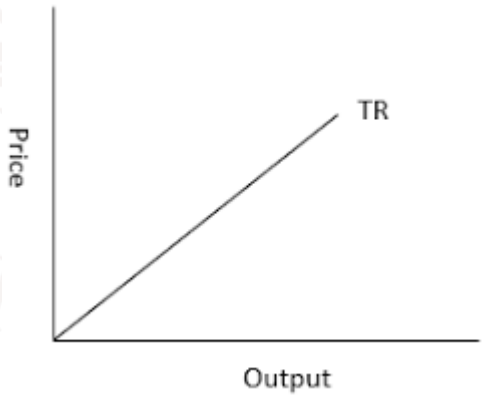


Figure 1: Graph for Total Revenue

Average revenue (AR)

Average revenue is the revenue per unit of the commodity sold. It can be obtained by dividing the TR by the number of units sold. Then AR = TR/Q, e.g. If TR is 150 by selling 10 units, AR = 150/15= 10.

Thus, average revenue means price. Since the demand curve shows the relationship between price and the quantity demanded, it also represents the average revenue or price at which the various amounts of a commodity are sold, because the price offered by the buyer is the revenue from the seller's point of view. Therefore, the average revenue curve of the firm is the same as the demand curve of the consumer.

Therefore, in economics, we use AR and price as synonymous except in the context of price discrimination by the seller. Mathematically, P = AR.

Marginal Revenue (MR)

Marginal revenue is the net increase in total revenue realized from selling one more unit of a product. It is the additional revenue earned by selling an additional unit of output by the seller.

Suppose a firm is selling 4 units of the output at the price of Rs.14 per unit. Now, if it wants to sell 5 units instead of 4 units and thereby the price of the product falls to Rs.12 per unit, then the marginal revenue will not be equal to Rs.12 at which the 5th unit is sold. 4 units, which were sold at the price of Rs.14 before, will all have to be sold at the reduced price of Rs.12 and that will mean the loss of 2 rupees on each of the previous 4 units. The total loss on the previous units will be equal to Rs. 8. Therefore, this loss of 8 rupees should be deducted from the price of Rs. 12 of the 5th unit while calculating the marginal revenue. The marginal revenue in this case, therefore, will be Rs. 12 – Rs. 8 = Rs. 4 and not Rs. 12 which is the average revenue.

Marginal revenue can also be directly calculated by finding out the difference between the total revenue before and after selling the additional unit of the product.

Total revenue when 4 units are sold at the price of Rs.14 = 4 X 14 = Rs. 56

Total revenue when 5 units are sold at the price of Rs.12 = 5 X 12 = Rs. 60

Therefore, Marginal revenue or the net revenue earned by the 5th unit = 60- 56 = Rs.4.

Thus, Marginal revenue of the nth unit = difference in total revenue in increasing the sale from n-1 to n units or Marginal revenue = price of nth unit minus loss in revenue on previous units resulting from price reduction.

The concept is important in microeconomics because a firm's optimal output (most profitable output) is where its marginal revenue equals its marginal cost i.e., as long as the extra revenue from selling one more unit is greater than the extra cost of making it, it is profitable to do so.

It is usual for marginal revenue to fall as output goes up both at the level of a firm and that of a market because lower prices are needed to achieve higher sales or demand respectively.

MR = $\frac{\Delta TR}{\Delta Q}$

= where TR represents change in TR

And Q indicates change in total quantity sold. Also, MR = TRn – TRn-1

Marginal revenue is equal to the change in total revenue over the change in quantity.

Marginal Revenue = (Change in total revenue) divided by (Change in sales) Table 8.1 represents the price – revenue sheet.

Table 1 represents the price – revenue sheet.

Table 1: Price Revenue Sheet

Units	**Price**	**TR**	**AR**	**MR**
1	20	20	20	-
2	18	36	18	16
3	16	48	16	12
4	14	56	14	8
5	12	60	12	4

There is another way to see why marginal revenue will be less than price when a demand curve slopes downward. Price is average revenue. If the firm sells 4 units for Rs.14, the average revenue for each unit is Rs.14.00. But as a seller sells more, the average revenue (or price) drops, and this can only happen if the marginal revenue is below price, pulling the average revenue down.

If one knows marginal revenue, one can tell what happens to total revenue if sales change. If selling another unit increases total revenue, the marginal revenue must be greater than zero. If marginal revenue is less than zero, then selling another unit reduces total revenue. If marginal revenue is zero, then selling another unit does not change total revenue. This relationship exists because marginal revenue measures the slope of the total revenue curve.

Relationship between total revenue, average revenue and marginal revenue concepts

In order to understand the relationship between TR, AR and MR, we can prepare a hypothetical revenue schedule. Table 2 represents a hypothetical revenue schedule that shows the relationship between TR, AR and MR.

Table 2: Hypothetical Revenue Schedule

Number of Units sold	TR (Rs.)	AR (Rs.)	MR (Rs.)
1	10	10	10
2	18	9	8
3	24	8	6
4	28	7	4
5	30	6	2
6	30	5	0
7	28	4	-2

From the table, it is clear that:

1. MR falls as more units are sold.
2. TR increases as more units are sold but at a diminishing rate.
3. TR is the highest when MR is zero.
4. TR falls when MR becomes negative.
5. AR and MR both fall, but the fall in MR is greater than AR i.e., MR falls more steeply than AR.

Thus, we have discussed the relationship between total revenue, average revenue and marginal revenue concept.

Relationship between AR and MR and the nature of AR and MR curves under different market conditions

Now, we will discuss the relationship between AR and MR and the nature of AR and MR curves under different market conditions.

Under perfect market

Under perfect competition, an individual firm by its own action cannot influence the market price. The market price is determined by the interaction between demand and supply forces. A firm can sell any amount of goods at the existing market prices. Hence, the TR of the firm would increase proportionately with the output offered for sale. When the total revenue increases in direct proportion to the sale of output, the AR would remain constant. As the good's market price is constant without any variation due to changes in the units sold by the individual firm, the extra output would fetch proportionate increase in the revenue. Hence, MR and AR will be equal to each other and remain constant. This will be equal to the price.

Table 3 shows the revenue comparison chart under perfect market conditions. Figure 2 shows a graphical representation of the data in table 2.

Table 3: Revenue Comparison Chart under Perfect Market

Number of Units sold	**Price per Unit Rs. 8.00**		
	AR	**TR**	**MR**
1	8	8	8
2	8	16	8
3	8	24	8
4	8	32	8
5	8	40	8
6	8	48	8

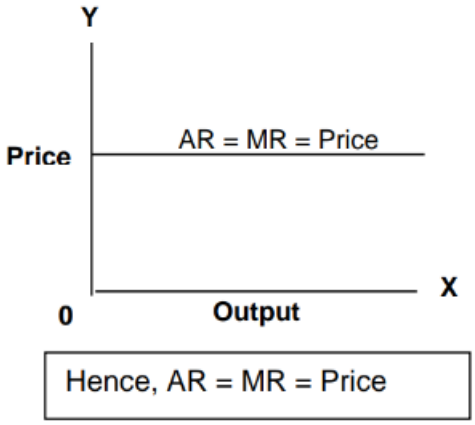


Figure 2: Price Graph under perfect market

Under perfect market conditions, the AR curve will be a horizontal straight line and parallel to OX axis. This is because a firm has to sell its product at the constant existing market price. The MR curve also coincides with the AR curve. This is because additional units are sold at the same constant price in the market.

Under imperfect market

Under all forms of imperfect markets, the relation between TR, AR, and MR is different. This can be understood with the help of the following imaginary revenue schedule.

Table 4 shows the revenue comparison chart under imperfect market conditions. Figure 3 shows a graphical representation of the data in table 4

Table 4: Revenue Comparison Chart under Imperfect Market

Number of Units sold	TR	AR or Price inRs.	MR
1	10	10	10
2	18	9	8
3	24	8	6
4	28	7	4
5	30	6	2
6	30	5	0

7	28	4	-2

From table 4, it is clear that:

In order to increase sales, a firm has to reduce its price, hence AR falls.

1. As a result of the fall in price, TR increases but at a diminishing rate.
2. TR will be higher when MR is zero
3. TR falls when MR becomes negative
4. Both AR (Average Revenue) and MR (Marginal Revenue) decline, but the fall in MR will be greater than the fall in AR (Average Revenue).

The relationship between AR and MR curves is determined by the elasticity of demand on the average revenue curve

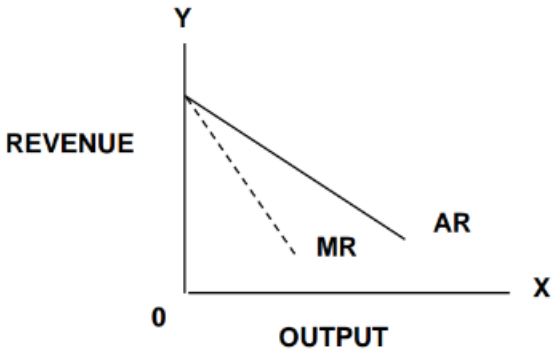


Figure 3: Revenue Graph under Imperfect Market

Under imperfect market: conditions, the AR curve of an individual firm slopes downward from left to right. This is because a firm can sell larger quantities only when it reduces the price. Hence, the AR curve has a negative slope.

The MR curve is similar to that of the AR curve, but MR is less than AR. AR and MR curves are different. Generally, the MR curve lies below the AR curve.

The AR curve of the firm or the seller and the demand curve of the buyer are similar.

Because the demand curve graphically represents the quantities demanded by the buyers at various prices, it shows the AR at which the various amounts of goods are sold by the seller. This is because the price paid by the buyer is the revenue for the seller (One man's expenditure is another man's

income). Hence, the AR curve of the firm is the same thing as that of the demand curve of the consumers. Figure 8.4 depicts the relationship between the AR curve and the demand curve.

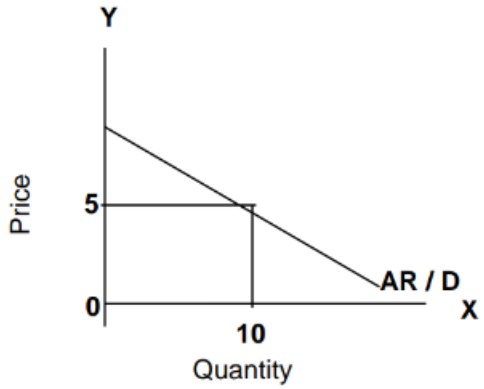


Figure 4: Relationship Between AR Curve and Demand Curve

Suppose that a consumer buys 10 units of a product, when the price of the product is Rs. 5 per unit. Hence, the total expenditure is 10 x 5 = Rs. 50/-. The seller is selling 10 units at the rate of Rs.5 per unit. Hence, his total income is 10 x 5 = Rs. 50/-. Thus, it is clear that AR curve and demand curve are really one and the same.

3. RELATIONSHIP BETWEEN REVENUES AND PRICE ELASTICITY OF DEMAND

In this section, we will discuss the relationship between revenues and price elasticity of demand.

Elasticity of demand, average revenue and marginal revenue

A very useful relationship exists between elasticity of demand, average revenue and marginal revenue at any level of output. The elasticity of demand at any point on a consumer's demand curve is the same as the elasticity on the given point on the firm's average revenue curve. With the help of the point elasticity of demand, we can study the relationship between average revenue, marginal revenue and elasticity of demand at any level of output. Figure 5 depicts the effect of elasticity of demand on AR and MR curves.

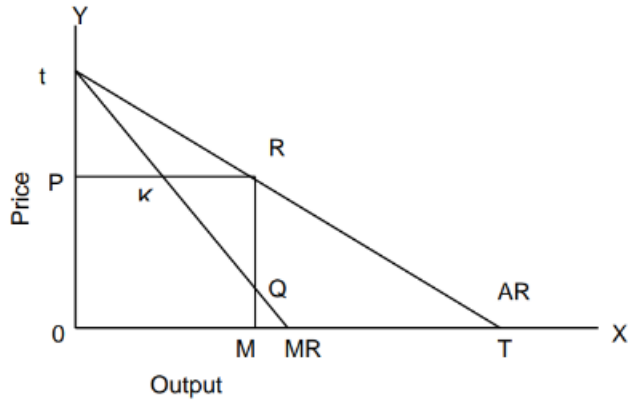


Figure 5: Effect of Elasticity of Demand on AR and MR Curves

In the diagram, AR and MR respectively are the average revenue and the marginal revenue curves. Elasticity of demand at point R on the average revenue curve = RT/Rt. Now, in the triangles PtR and MRT

tPR = RMT (right angles)

tRP = RTM (corresponding angles) PtR = MRT (being the third angle)

Therefore, triangles PtR and MRT are equiangular.

Hence RT / Rt = RM / tP

In the triangles PtK and KRQ PK = RK

PKt = RKQ (vertically opposite) tPK = KRQ (right angles)

Therefore, triangles PtK and RQK are congruent (i.e., equal in all respects).

Hence Pt = RQ

Elasticity at R = RT / Rt = RM / tP = RM / RQ

It is clear from the diagram that

RM - QM = RQ

Hence, elasticity at R = RM / RM – QM

It is also clear from figure 7.5 that RM is average revenue and QM is the marginal revenue at the output OM which corresponds to the point R on the average revenue curve.

Therefore, elasticity at R = Average Revenue / Average Revenue – Marginal Revenue

If A stands for average revenue, M stands for marginal revenue and e stands for point elasticity on the average revenue curve then, e = A / A – M.

Thus, elasticity of demand is equal to AR divided by AR minus MR.

By using the above elasticity formula, we can derive the formula for AR and MR separately.

eA – eM = A; bringing all 'A' together, we have eA – A = eM

A (e – 1) = eM

A = eM / e – 1 A =M (e / e –1)

Therefore, Average Revenue or price = M (e / e – 1)

Thus, the price per unit (i.e., AR) is equal to marginal revenue x elasticity over elasticity minus one. The marginal revenue formula can be written straight away as

M = A ((e – 1) / e)

The general rule therefore is: at any output,

Average Revenue = Marginal Revenue x (e / e – 1) and Marginal Revenue = Average Revenue x (e – 1 / e)

Where, e stands for point elasticity of demand on the average revenue curve. With the help of these formulae, we can find marginal revenue at any point from average revenue at the same point, provided we know the point elasticity of demand on the average revenue curve. Suppose that the price of a product is Rs.8 and the elasticity is 4 at that price. Marginal revenue will be:

M = A ((e – 1) / e)

= 8 ((4 – 1) / 4)

= 8 x 3 /4

= 24 / 4

= 6. Marginal Revenue is Rs. 6.

Suppose that the price of a product is Rs. 4 and the elasticity coefficient is 2 then the corresponding MR will be:

M = A ((e-1) / e)

= 4 ((2 – 1) / 4)

= 4 x 1 / 4

= 4 / 4

= 1. Marginal revenue is Rs. 1

Suppose that the price of commodity is Rs.10 and the elasticity coefficient at that price is 1, MR will be:

M = A ((e-1) / e)

=10 ((1-1) /1)

=10 x 0/1

= 0

Whenever elasticity of demand is unity, marginal revenue will be zero, whatever may be the price (or AR). It follows from this that if a demand curve shows unitary elasticity throughout its length, the corresponding marginal revenue will be zero throughout, i.e., the X axis itself will be the marginal revenue curve.

Thus, the higher the elasticity coefficient, the closer is the MR to AR or price. When elasticity coefficient is one for any given price, the corresponding marginal revenue will be zero; marginal revenue is always positive when the elasticity coefficient is greater than one and, marginal revenue is always negative when the elasticity coefficient is less than one.

Kinked demand curve and the corresponding marginal revenue curve

Figure 6 depicts a kinked demand curve corresponding to the marginal revenue curve.

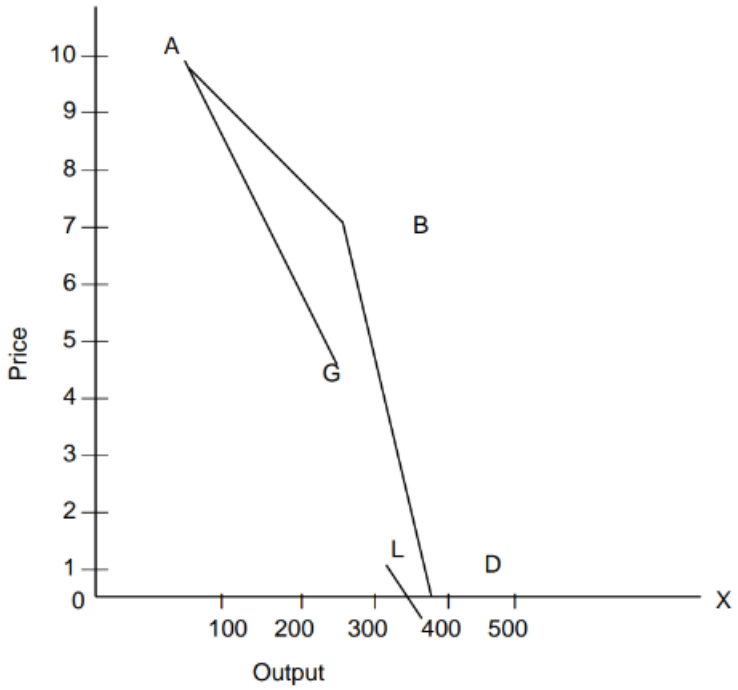


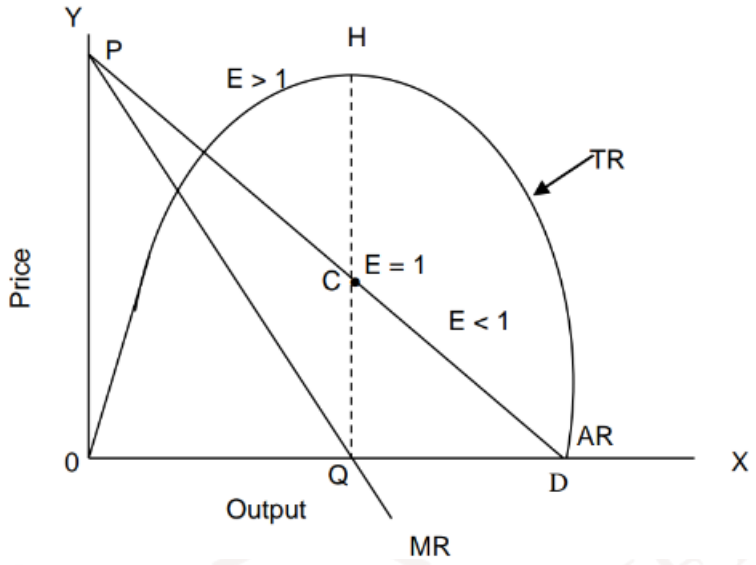
Figure 6: Kinked Demand Curve

We measure quantity on the X axis and price on the Y axis. The demand curve AD has a kink at point B, thus exhibiting two different characteristics.

From A to B, it is elastic but from B to D it is inelastic. Because the demand is elastic from A to B. a very small fall in price causes a very big rise in demand, but to realize the same increase in demand, a very big fall in price is required as the demand curve assumes inelastic shape after point B. The corresponding marginal revenue curve initially falls smoothly, though at a greater rate. In figure 6 there is a gap in MR between output 300 and 350.

Generally, an oligopolistic firm that faces a kinked demand curve will make a good gain when it reduces the price a little before the kink (point B), but if it lowers the price below B; the rival firms will lower their prices too; accordingly, the price cutting firm will not be able to increase its sales correspondingly or may not be able to increase its sales at all. As a result, the demand curve of price cutting firm below B is more inelastic. The corresponding MR curve is not smooth but has a gap or discontinuity between G and L. In certain cases, the kinked demand curve may show a high elasticity in the lower portion of the demand curve beyond the kink and low elasticity in higher portion of the demand curve before the kink. Marginal revenue in such a demand curve will show a gap but instead of at a lower level, it will start at a higher level. Figure 7 depicts the relationship between AR, MR, TR and elasticity of demand.

Figure 7: Relationship Between AR, MR, TR and Elasticity of Demand



In figure 7, AR is the average revenue curve, MR is the marginal revenue curve and OD is the total revenue curve. At the middle point C of the average revenue curve, elasticity is equal to one. On its lower half, it is less than one and on the upper half, it is greater than one. MR corresponding to the middle point C of the AR curve is zero. This is shown by the fact that MR curve cuts the x axis at Q which corresponds to the point C on the AR curve. If the quantity is greater than OQ it will correspond to that portion of the AR curve where e<1 and marginal revenue is negative because MR goes below the X axis. Similarly, for a quantity less than OQ, e>1 and the marginal revenue is positive. This means that if quantity greater than OQ is sold, the total revenue will diminish and for a quantity less than OQ, the total revenue TR will increase. Thus, the total revenue TR will be at its maximum at point H, where elasticity is equal to one and marginal revenue is zero.

Significance of revenue curves

The relationship between price elasticity of demand and total revenue is important because every firm has to decide whether to increase or decrease the price depending on the price elasticity of demand of the product. If the price elasticity of demand for the firm's product is relatively elastic, it will be advantageous to reduce price as it increases the firm's total revenue. On the other hand, if the price elasticity of demand for the firm's product is relatively inelastic, it should raise the price as it increases total revenue.

Average revenue, which is the price per unit, when considered along with average cost will show the firm whether it is profitable to produce and sell. If average revenue is greater than average cost, the firm is getting excess profit; if it is less than average cost, the firm is running at a loss.

Firm's profit is at its maximum at a point where Marginal revenue is equal to Marginal cost. Any increase in output beyond that point will mean a loss of additional units produced; restriction of output before that point will mean lower profit. Thus, the concept of average revenue is relevant to find out whether the firm is running on profit or loss; the concept of marginal revenue together with marginal cost will show profit maximizing output for the firm.

SELF-ASSESSMENT QUESTIONS – 1

Fill in the blanks:	
1	_____is the total income realized from the sale of its output at a price.
2	TR / Q = _____.
3	Additional revenue earned by selling an additional unit of output is called _____.
4	The AR curve coincides with the MR curve and runs parallel to the OX axis under _____ competition.
5	AR and MR curve slope downwards under the condition of _____ competition.
True or False Questions:	
6	Total revenue is maximum when marginal revenue is the highest. (True/False)
7	TR increases when price is increased if price elasticity of demand is greater than 1. (True/False)

8	The AR curve of the firm or the seller and the demand curve of the buyer are similar. (True/False)
9	With the increase in Q, AR and MR both fall, but the fall in MR is greater than AR i.e., MR falls more steeply than AR. (True/False)
10	MR and AR behave similarly with change in prices and output. (True/False)

4. PRICING POLICIES

In this section, we will discuss the pricing policies. A detailed study of the market structure gives us information about the way in which prices are determined under different market conditions. However, in reality, a firm adopts different policies and methods to fix the price of its products. Pricing policy refers to the policy of setting the price of the product or products and services by the management after taking into account various internal and external factors, forces and its own business objectives. Pricing Policy basically depends on price theory that is the cornerstone of economic theory. Pricing is considered as one of the basic and central problems of economic theory in a modern economy. Fixing prices is the most important function of managerial decision making because market prices are changed by the company, affects the present and future production plans, pattern of distribution, nature of marketing etc. Above all, the sales revenue and profit ratio of the producer directly depend upon the prices. Hence, a firm has to charge the most appropriate price to the customers. Charging an ideal price, which is neither too high nor too low, would depend on a number of factors and forces. There are no standard formulae or equations in economics to fix the best possible price for a product. The dynamic nature of the economy forces a firm to raise and reduce the prices continuously. Hence, prices fluctuate over a period of time.

Generally speaking, in economic theory, we take into account only two parties, i.e., buyers and sellers, while fixing the prices. However, in practice, many parties are associated with pricing of a product. They are rival competitors, potential rivals, middlemen, wholesalers, retailers, commission agents and above all the Government. Hence, we should give due consideration to the influence exerted by these parties in the process of price determination.

Broadly speaking, the various factors and forces that affect the price are divided into two categories. They are as follows:

I. External Factors – There are several external factors that affect the price. They are as follows:

1. Demand, supply and their determinants
2. Elasticity of demand and supply
3. Size of the market
4. Goodwill, name, fame and reputation of a firm in the market
5. Purchasing power of the buyers

6. Buyers' behavior in respect of particular product
7. Availability of substitutes and complements
8. Government's policy relating to various kinds of incentives, disincentives, controls, restrictions and regulations, licensing, taxation, export and import, foreign aid, foreign capital, foreign technology
9. Competitor's pricing policy
10. Social considerations

II. Internal Factors – There are several internal factors that affect the price. They are as follows:

1. Objectives of the firm
2. Production Costs
3. Quality of the product and its characteristics
4. Scale of production
5. Efficient management of resources
6. Policy towards percentage of profits and dividend distribution
7. Advertising and sales promotion policies
8. Wage policy and sales turnover policy, etc.
9. The stages of the product on the product lifecycle
10. Use pattern of the product
11. Extent of the distinctiveness of the product and extent of product differentiation practiced by the firm
12. Composition of the product and life of the firm

Thus, multiple factors and forces affect the pricing policy of a firm.

5. OBJECTIVES OF PRICING POLICIES

In this section, we will discuss the objectives of pricing policies. While formulating its pricing policy, a firm has to consider various economic, social, political and other factors. The following objectives are to be considered while fixing the prices of the product:

1. ***Profit maximisation in the short term*** **–** The primary objective of the firm is to maximise its profits. Pricing policy as an instrument to achieve this objective should be formulated in such a way as to maximise sales revenue and profit. Maximum profit refers to the highest possible profit. In the short run, a firm not only should be able to recover its total costs, but also should get excess revenue over costs. It may follow a skimming price policy, i.e., charging a very high price when the product is launched to cater to the needs of only a few sections of people. Alternatively, it may adopt a penetration pricing policy i.e., charging a relatively lower price in the later stages in the long run so as to attract more customers and capture the market.
2. ***Profit optimisation in the long run*** **–** The traditional profit maximisation hypothesis may not prove beneficial in the long run. Optimum profit refers to the most ideal or desirable level of profit. Hence, earning the most reasonable or optimum profit has become a part and parcel of a sound pricing policy of a firm in recent years.
3. ***Price stabilisation*** **–** Price stabilisation over a period of time is another objective. A stable price policy can only win the confidence of customers and may add to the goodwill of the concern. It builds up the reputation and image of the firm.
4. ***Preventing entry of competition*** **–** Firms try to match the prices of their products with those of their rivals to expand the volume of their business. Most of the firms are not merely interested in meeting competition but are keen to prevent it.
5. ***Maintenance of market share*** **–** Market share refers to the share of a firm's sales of a particular product in the total sales of all firms in the market. The economic strength and success of a firm is measured in terms of its market share. Hence, the pricing policy has to assist a firm to maintain its market share.
6. ***Capturing the market*** **–** Another objective in recent years has been to capture the market, dominate the market and command and control the market in the long run. In order to achieve this goal, sometimes, the firm fixes a lower price for its product and at other times, it may even sell at a loss in the short term. It may prove beneficial in the long run. Such pricing is generally followed in price sensitive markets.

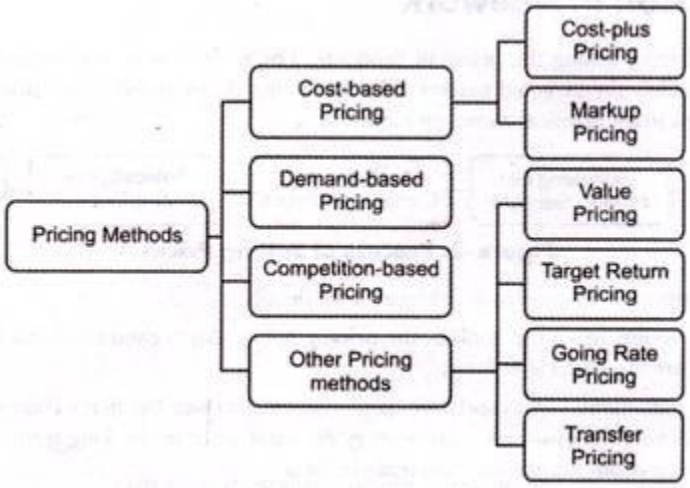
7. ***Entry into new markets*** **–** Apart from growth, market share expansion, diversification in its activities, firms make special attempts to enter into new markets. The price set by a firm has to be so attractive that the buyers in other markets have to switch over to the products of the candidate firm.
8. ***Deeper penetration of the market*** **–** The pricing policy has to be designed in such a manner that a firm can make inroads into the market with minimum difficulties. Deeper penetration is the first step in the direction of capturing and dominating the market in the later stages.
9. ***Achieving a target return*** **–** A predetermined target return on capital investment and sales turnover is another long run pricing objective of a firm. The targets are set according to the position of the individual firm. Hence, prices of the products are calculated so as to earn the target return on cost of production, sales and capital investment. Different target returns may be fixed for different products or brands or markets but such returns should be related to a single overall rate of return target.
10. ***Target profit on the entire product line irrespective of profit level of individual products*** **–** The price set by a firm should increase the sales of all the products rather than yield a profit on one product only. A rational pricing policy should always keep in view the entire product line and maximum total sales revenue from the sale of all products. A product line may be defined as a group of products which have similar physical features and perform generally similar functions. In a product line, a few products are regarded as less profit earning products and others are considered as more profit earning. Hence, a proper balance in pricing is required.
11. ***Long run welfare of the firm*** **–** A firm has multiple objectives. They are laid down on the basis of past experience and future expectations. Simultaneous achievement of all objectives is necessary for the overall growth of a firm. The objective of the pricing policy has to be designed in such a way as to fulfil the long run interests of the firm keeping internal conditions and external environment in mind.
12. ***Ability to pay*** **–** Pricing decisions are sometimes taken on the basis of the ability to pay of the customers, i.e., a higher price can be charged to those who can afford to pay. Such a policy is generally followed by those people who supply different types of services to their customers.
13. ***Ethical pricing*** **–** Basically, pricing policy should be based on certain ethical principles. Business without ethics is a sin. While setting the prices, some moral standards are to be followed. Although profit is one of the most important objectives, a firm cannot earn it in a moral vacuum. Instead of squeezing customers, a firm has to charge moderate prices for its

products. The pricing policy has to secure reasonable amount of profits to a firm to preserve the interests of the community and promote its welfare.

Besides these goals, there are various other objectives such as promotion of new items, steady working of plants, maintenance of comfortable liquidity position, making quick money, maintaining regular income to the company, continued survival, rapid growth of the firm etc., which firms may consider while taking pricing decisions.

6. PRICING METHODS

In this section, we will discuss the pricing methods.



Cost-based pricing can be defined as a pricing method in which a certain percentage of the total cost of production is added to the cost of the product to determine its selling price. Cost-based pricing can be of two types, namely,

I. cost-plus pricing

- Rate of return pricing

II. markup pricing.

I. Full – cost pricing or cost - plus pricing method

In the cost-plus pricing method, a fixed percentage, also called mark-up percentage, of the total cost (as a profit) is added to the total cost to set the price. For example, the XYZ organization bears the total cost of Rs. 100 per unit for producing a product. It adds Rs. 50 per unit to the price of product as' profit. In such a case, the final price of a product of the organization would be Rs. 150.

P = AVC + AVC (M)

AVC= Average Variable Cost

M = Mark-up percentage

AVC (m) = Gross profit margin

Mark-up percentage (M) is fixed in which AFC and net profit margin (NPM) are covered.

AVC (m) = AFC+ NPM

ii. For determining average variable cost, the first step is to fix prices. This is done by estimating the volume of the output for a given period of time. The planned output or normal level of production is taken into account to estimate the output.

The second step is to calculate the Total Variable Cost (TVC) of the output. TVC includes direct costs, such as cost incurred in labor, electricity, and transportation. Once TVC is calculated, AVC is obtained by dividing TVC by output, Q. [AVC= TVC/Q]. The price is then fixed by adding the mark-up of some percentage of AVC to the profit [P = AVC + AVC (m)].

Full cost pricing is one of the simplest and most common methods of pricing adopted by different firms. Cost refers to fully allocated costs. According to Joel Dean, it has three components –

i) Actual cost which refers to the actual or total expenses incurred in production, for e.g., wage bills, raw material cost, overhead charges, etc.

ii) Expected cost which refers to the forecast for the pricing period on the basis of expected prices, output rate and productivity.

iii) Standard cost which refers to cost incurred at the normal level of output.

In brief, a firm computes the selling price of its product by adding a certain percentage to the average total cost of the product. The percentage added to costs is called margin or mark-ups. Hence, this method is also called margin – pricing and mark – up pricing.

Cost-plus Pricing = Cost + Fair Profit

Fair profit means a fixed percentage of profit mark-ups. It is arbitrarily determined. The margin of profits included in the price of a product differs from industry to industry and from commodity to commodity on account of differences in competitive strength, cost of production, total turnover, accounting practices, etc. Past traditions, directives from trade associations and, guidelines from the government may also decide the percentage of profits.

This method envisages covering the total costs incurred in producing and selling a commodity. In this case, businessmen do not seek supernormal profits. Hence, a price based on full average cost is the

'right cost', the one which ought to be charged based on the idea of fairness under oligopoly and monopolistic competition.

Illustration

Production = 8000 units.

Total Fixed Cost = Rs. 30,000

Total Variable Cost = Rs. 50,000

Total Cost = Rs. 80,000

Per Unit Cost = 80,000 / 8000 = Rs. 10

Now consider 20% as Net Profit Margin

On Cost = $^{20}/_{100}$ X 10 = 2 INR

Cost Price = 10 INR

20% NPM on Cost = + Rs. 2

Selling Price = Rs. 12

Evaluation of the full cost pricing method

Generally, the firms will not have information about demand conditions, nature and degree of competition, technology used, etc. Further, modern business conditions are extremely uncertain. Besides, a firm may be producing or selling innumerable varieties of goods and to calculate prices on the basis of profit maximization may be almost impossible. The cost-plus method is convenient, as the firms have to only add some standard mark-up to their cost. Over a period of time, through trial and error, they can find out the proper mark–up. The supreme merit of this method lies in its mechanical simplicity and its apparent fairness.

It is safer, cheaper and imparts competitive stability particularly when there is tough competition in the market. It is useful particularly in product tailoring and public utility pricing. It is justified on moral grounds because price based on costs is a just price.

According to Professor Joel Dean, it is the best method of pricing in the case of new products because if the firm is able to realize its normal profits, only then can it take a decision to produce and market a product.

This method attaches too much significance to allotted costs and mark- ups. It tends to diminish the interest of the producer in cost control. However, many firms adopt this method of pricing due to its inherent benefits.

- Rate of return pricing

Rate of return pricing is a modified form of full cost pricing. Under this method, a producer decides a predetermined target rate of return on capital invested. Full–cost pricing considers the mark-ups or profit arbitrarily. Instead of setting the percentage arbitrarily, a firm will determine the average mark-up on costs necessary to produce a desired rate of return on the company's investments. Thus, under this method, price is determined along a planned rate of return on investment. In this case, a company estimates future sales, future costs and arrives at a mark–up that will achieve a target return on a company's investment.

Professor Davies and Hughes in their book, "Managerial Economics" have used the following formula to calculate the desired rate of return when a mark-up is applied on cost.

Price=1−Desired Rate of Return Cost

Evaluation of rate of return pricing

As it is a refined version of cost-plus pricing, it is superior to cost plus pricing in the following two ways:

i) The analysis is based on standard cost which is computed on the basis of normal output; and,

ii) The profit mark-up is based on a planned rate of return on investment and not on any arbitrary figure.

As it is a refined form of cost-plus pricing method, all the merits and demerits of cost-plus pricing method apply to rate of return pricing too.

II. Markup pricing: Refers to a pricing method in which the fixed amount or the percentage of cost of the product is added to product's price to get the selling price of the product.

Percentage mark–up = Capital employed/ Total annual cost X Planned rate of return

Let us suppose that the capital employed by a firm is Rs. 16 lakhs and the total cost is Rs. 12 lakhs with a planned rate of return of 30 percent. By making use of the above formula, we can find out the percentage mark-up of the firm in the following way.

Illustration

Capital employed / Total annual cost X planned rate of return

= 16 / 12 X30% = 40

Production = 10,000 units

Total Cost = 4,00,000

Per Unit Cost = 4,00,000 / 10,000 = Rs. 40

30% of Rs. 40 is = 30/100X 40 = Rs 12

Now, if the total cost per unit is = 40

30% mark-up = 12

The selling price would be = 52

The mark-up is thus carefully planned and calculated, as different from the arbitrary percentage used in the cost-plus pricing.

Price = Total cost per unit + Mark-up

The management will regard this price as the base price applicable over a period of time. However, when the cost of production changes as a result of changes in the prices of raw materials or due to changes in the levels of wages, the management can change the price suitably. Besides, the base price can be modified suitably according to changes in demand and competitive conditions in the market.

Demand-based Pricing:

Demand-based pricing is a method of pricing in which the price of a product is fixed based on the demand. A higher price is charged when the demand for the product is greater and here the organization sets a very high price to make profits when the demand and sales are high; whereas, if the demand for a product is less, a low price is charged to attract customers.

Demand-based pricing will work only if the demand in the market is analyzed correctly. This type of pricing is common in hospitality, travel and tourism. For instance, airlines, hotels and travel agents lower their prices or charges when the demand is low during off seasons in comparison to periods of high demand such as holiday seasons.

Competition-based Pricing:

Competition-based pricing is a method where the product price is fixed based on the prices of competitors' products. The price may be fixed ar a higher, lower, or equal price based on the competitor's product price and the pricing strategy of the business.

Other Pricing Methods:

Value Pricing:

When a business entity tries to create customer loyalty and attract a large number of customers by charging low prices and in turn provide high- quality products. The business entity creates an image of a low-cost producer who never compromises on quality and delivers high standard products. This can be done by improving its research and development process. Value pricing is also called value-optimized pricing.

Target Return Pricing:

The estimated profit or the rate of return on investment is first calculated and then the product price is fixed based on this. The price of a product is fixed on the basis of expected profit.

Going rate pricing

Going rate pricing is the opposite of full cost pricing. In this method, emphasis is given to market conditions rather than to costs. Generally, we come across this method of pricing under oligopoly market especially underprice leadership. Under this method, a firm fixes its price according to the price fixed by the leader. A firm has monopoly power over the product it produces and can charge its own price and face all the consequences of

monopoly. However, a firm chooses the price which is going on the market and charges a particular price that the other followers are charging.

This type of pricing is not the same as accepting a price that is set in a perfectly competitive market. A firm has some power to fix the price but instead of doing so, it adjusts its own price to the general price structure in the industry. Hence, this method of pricing is known as acceptance pricing. Normally, under this method, the industry tries to determine the lowest prices that the sellers or followers can afford to accept considering various alternatives.

The price follower firms, however, compare the price of the leader and the leader firm's cost-revenue conditions and long run profitability. As small firms recognize the big firm as their leader, they try to imitate their leader in pricing decisions. Since a price leader is a firm with a successful profit history, significant market share and long experience in market matters, the imitating firms follow the leader in the hope of earning larger profits under the shelter of the leader's price umbrella.

Imitation is the easy way of decision making. The follower uses another firm's market analysis without worrying about demand and cost estimates. Many executives desire to devote minimum time to pricing decisions and hence, they follow this method.

This policy is not confined to only small business firms. Even large firms follow a price set by a price leader or by the market. Some firms adjust their costs to a predetermined price by keeping their costs within certain percentage limits of their selling prices in order to achieve the targeted profit. This policy suits those products which have reached a mature stage and where both customers and rivals have come to accept a stable price. The going rate pricing is generally adopted when:

a. Costs are difficult to measure
b. The firm wants to avoid tension of price rivalry in the market
c. There is price leadership of a dominant firm in the market

This method of pricing is easy to adopt, economical and rational. It helps in avoiding cutthroat competition among firms.

'Imitation pricing it' is a variant of going rate pricing. The firms which join the industry late just imitate the price fixed by the leader. This is the same as going rate pricing.

Transfer Pricing:

Involves selling of goods and services within various departments of the organization to manage the profit and loss ratios of different departments. One department of an organization can sell its products to another department at low prices. Sometimes, transfer pricing is used to show higher profits in the organization by showing fake sales of products within departments.

Administered prices

The term administered prices was introduced by Keynes for the prices charged by a monopolist and therefore it is determined by considerations other than marginal cost. A monopolist being a price maker consciously administers the price of his product.

Indian economists like L.K. Jha and Malcolm Adiseshaiah, however, have a slightly different conception about administered prices. According to the Indian economists, an administered price for a commodity is the one which is decided and arbitrarily fixed by the government. It is not allowed to be determined by the free play of market forces of demand and supply. In short, administered prices are the prices which are fixed and enforced by the government in the overall interest of the economy.

Administered prices are fixed by the government for a few carefully selected goods like steel, coal, aluminum, fertilizers, petroleum, cooking gas, etc.

These products are the raw materials for other industries and there is a great need for establishing and stabilizing the total output and their prices. The public distribution system is also subject to administered prices.

Administered prices are normally set on the basis of cost plus a stipulated margin of profit. They represent a pool price where individuals producing units are being granted retention prices. These retention prices may either be uniform or different for different units. As cost of production changes, administered prices also would be modified. This is the right method of pricing and it is based on logical considerations.

Objectives of administered pricing

The main objectives of administered pricing are as follows:

1. To protect the interests of weaker sections against high prices
2. To curb or to encourage the consumption of certain commodities
3. To contain inflation and ensure price stability
4. To counter stagflation and the consequent recession
5. To mobilize revenue for the government
6. To ensure efficient allocation of resources among different users
7. To improve living standards of the masses and promote their economic welfare
8. To ensure equitable distribution of certain goods which are scarce in supply
9. To achieve macroeconomic goals like welfare, equity and stability

It is quite clear from the above discussion that administered prices have certain defects. In order to make administered prices more realistic, they should reflect the cost realities from time to time and quickly respond to these changes in the most pragmatic manner. The government administration should be active and prompt at the decision-making level. This will bring a reputation to administered prices and they may be accepted without much criticism.

Marginal Cost Pricing

It is based on a purely economic concept of equilibrium of a firm, where marginal cost is equal to marginal revenue. Under this method, price is determined on the basis of marginal cost, which refers to the cost of producing additional units. Price based on marginal cost will be much more aggressive than the one based on total cost. A firm with large unused capacity will have to explore the possibility of producing and selling more. If the price is sufficient to cover the marginal cost, particularly in times of recession, the firm should be able to produce and sell the commodity and can think of recovering the total cost in the long run.

Although this method sounds excellent theoretically, it has the serious limitation of ascertaining the marginal cost.

SELF-ASSESSMENT QUESTIONS – 2

Fill in the blanks:	
11	Cost plus pricing = cost + _____ .
12	We come across going rate pricing generally under _____ market.
13	The objective of charging high prices for new products is to _____ from the market.
14	The rate of return pricing = Total cost per unit + _____ .
15	Administered prices are the prices which are fixed and enforced by the _____ in the overall interest of the community.
True or False Questions:	
16	When an individual firm increases its production in a competitive market, it is likely that the market price will fall. (True/False)
17	For a firm in a competitive market, marginal revenue is always equal to average revenue. (True/False)
18	A profit-maximizing firm in a competitive market will increase production when average revenue exceeds marginal cost. (True/False)

19	After comparing the marginal revenue and marginal cost from each unit produced, a firm in a competitive market can determine the profit-maximizing the level of production. (True/False)
20	When a firm in a monopoly charges a higher price, fewer of its goods are sold. (True/False)

7. SUMMARY

Let us recapitulate the important concepts discussed in this unit:

- Knowledge of cost and revenue concepts is of very great importance in understanding the various methods of price-output determination and pricing policies under both perfect and imperfect markets.
- Total revenue refers to the total receipts from the sale of the goods, Average revenue refers to the revenue per unit of the commodity sold and marginal revenue is the additional revenue earned by selling an additional unit of output.
- The relationship between revenue and price elasticity of demand has practical significance in real business life. These two concepts help the management to take the right decision with regard to the size of the output and the determination of price.
- Different pricing policies and methods give an insight into the actual functioning of a firm. Dynamic conditions of the market necessitate frequent changes in the pricing policies and methods followed by a firm. While formulating its pricing policy a firm has to keep in its view some of the external factors like elasticity of demand, size of the market, government policy, etc., and internal factors like production costs, the stages of the product on the product life cycle etc.
- There are a few considerations to be kept in mind like the objectives of a firm, competitive situation in the market, cost of production, elasticity of demand, economic environment, government policy etc.
- The main objectives of the pricing policy are profit maximization, price stabilization, facing competitive situation, capturing the market etc. Market price of a product depends upon a number of factors like production cost, demand, consumer psychology, profit policy of the management, government policy, etc.
- There are different methods of pricing for both established products as well as new products. Full cost pricing or cost-plus pricing, is one of the simplest and most common methods of pricing adopted by different firms. Here the price is determined by adding a certain mark-up to the average total cost.

- Rate of return pricing is a modified form of full cost pricing where the mark-up is decided on the basis of capital employed. Going rate pricing is the opposite of full cost pricing generally followed under oligopoly market.

Here the firm just follows the price prevailing in the market without bothering about other things.

- Imitative pricing is similar to going rate pricing. Marginal cost pricing, where price is determined on the basis of marginal cost, is more theoretical than practical. Administered prices are the prices statutorily determined by the government for certain important goods like steel, cement etc.
- There are two schemes of pricing for a new product viz., skimming price and penetration price. Based on the market conditions and the cost conditions, these two methods are adopted.

8. GLOSSARY

Administered prices	-	Prices which are fixed and enforced by the government in the overall interest of the economy.
Average revenue	-	Revenue per unit of the commodity sold.
Full	-	cost pricing or cost - plus pricing: Firm computes the selling price of its product by adding a certain percentage to the average total cost of the product.
Going rate pricing	-	Firm fixes its price according to the price fixed by the leader.
Marginal cost pricing	-	Price is determined on the basis of marginal cost which refers to the cost of producing additional units.
Marginal revenue	-	Additional revenue earned by selling an additional unit of output by the seller.
Market share	-	Share of a firm's sales of a particular product in the total sales of all firms in the market.
Optimum profit	-	Ideal or desirable level of profit.
Penetration price policy	-	Instead of setting a high price, the firm may set a low price for a new product by adding a low mark-up to the full cost.
Pricing policy	-	Policy of setting the price of the product or products and services by the management after taking into account various internal and external factors, forces and its own business objectives.
Product line	-	A group of products which have similar physical features and perform generally similar functions.
Rate of return pricing	-	Producer decides a predetermined target rate of return on capital invested.

Revenue	-	Amount of money which the firm receives by the sale of its output in the market.
Skimming price policy	-	The system of charging high prices for new products.

9. TERMINAL QUESTIONS

1. Explain in brief the relationship between TR, AR and MR under perfect market conditions.
2. What do you mean by pricing policy?
3. What are the objectives of pricing policies?
4. Explain the various methods of pricing.

10. ANSWERS

Self-Assessment Questions

1. Total revenue
2. AR
3. Marginal revenue
4. Perfect
5. Imperfect
6. False
7. False
8. True
9. True
10. False
11. Fair profits
12. Oligopoly market
13. Skim the cream from the market
14. Mark-up
15. Government
16. False
17. True
18. True
19. True
20. True

Terminal Questions Answers

Answer 1: Under perfect competition, a firm can sell any amount of goods at the existing market prices. Hence, the TR of the firm would increase proportionately with the output offered for sale. When the total revenue increases in direct proportion to the sale of output, the AR would remain constant. As the good's market price is constant without any variation due to changes in the units sold by the individual firm, the extra output would fetch a proportionate increase in the revenue. Hence, MR and AR will be equal to each other and remain constant. This will be equal to the price. Refer to section 2.

Answer 2: Pricing policy refers to the policy of setting the price of the product or products and services by the management after taking into account various internal and external factors, forces and its own business objectives. Refer to section 4.

Answer 3: The following objectives are to be considered while fixing the prices of the product: profit maximization in the short term, profit optimization in the long run, price stabilization, preventing entry of competition. Refer to section 5.

Answer 4: There are several methods of pricing such as full – cost pricing or cost - plus pricing method, rate of return pricing, going rate pricing, administered prices, marginal cost pricing, customary pricing etc. Refer to section 6.

11. CASE STUDY

Real Steel in the Making

During the last two decades, post-initial liberalisation of the national economy and the steel industry sector, India's steel production and corresponding domestic steel consumption, in tandem, has increased from around 16 mtpa to around 65 mtpa.

During the last 20 years, the Indian steel sector, in general, has become significantly cost competitive and increasingly quality competitive, globally. This growth phenomenon is directly linked to the establishment of a few relatively large integrated iron and steel plants, namely, Essar, Jindal Vijayanagar (now JSW Steel) and Ispat (now JSW Ispat). Each took front-end innovative technology routes, and triggered global competitiveness. Tata Steel (India), SAIL and Vizag Steel have also improved their performance and growth phenomenally.

Incidentally, unlike pre-liberalisation, when each of the integrated steel plants was provided with captive iron ore mines – the only domestic resource input for competitiveness of the steel business in India – post- liberalisation, no integrated steel plant was provided with captive iron ore mines. Incidentally, the international trade prices of high-quality iron ore and coking coal in particular, are disproportionately high, largely because of significant monopolisation of these resources by 3 to 4 multi-national conglomerates.

In India, coking coal quality is poor to very poor, and quantity is insignificant, compared to the expanding needs of our steel sector. This coal is largely owned by Coal India Ltd. and older steel companies, set up prior to liberalisation in India. All the steel plants situated away from this only, meagre, coking coal resource in India in the east have to import coking coal from Australia, CIS and USA.

Among the major steel plants established post-liberalisation, Essar and Ispat are largely linked with natural gas, and the price is controlled by the government, due to their location in western India. JSW Steel, the steel plant with the largest capacity in one location in India, is linked to 100 per cent coking and steam coal import.

The present iron ore crisis in Karnataka gets partly linked to this key issue. The differential between the selling price of iron ore and its cost of production is a few times. No commodity or bulk mined product, such as iron ore should have super gross profit margins usurping the significant part of value in the supply chain, making investments in further downstream units unviable and unattractive.

The differential, if any, gets harmonised by combination of market forces and government legislations from time to time. This precisely didn't happen, and it got perpetuated by the governments and bureaucracy for obvious reasons for the last eight years or so.

In Karnataka, while 'subsistence'-level iron ore supply is being resumed from the 'dumps', the reserve price for auction has been fixed, linked to export price. The issue which isn't getting focus is that the iron and steel industry, even the sponge iron sector, cannot sustain itself in Karnataka or elsewhere, unless the iron ore comes on cost-plus-reasonable gross margin basis.

The reserve price for the present iron ore dumps for auction to steel plants has been fixed at around Rs. 3,000 for high-grade ore, and was sold at a premium to reserve price, solely for the reason of acute shortage created in the past few months. Under total duress, these steel companies are buying. Media attention is getting focused on the government that is earning correspondingly more revenue and a great service to the nation is being performed. Simultaneously, the exchequer is losing much more in the form of excise duties, service tax, CST, and VAT on the loss of steel production, since these taxes amount to Rs. 6000 per tonne of steel, as against a small amount of incremental royalty on the higher iron ore price.

However, this will directly impact on the cost of production of steel in Karnataka. Correspondingly, it will reduce the competitiveness and stakeholders' value of all the steel companies in the region. If this duress pricing is established, then as and when some of the mines are allowed to resume their operation, after due correction of their environmental and other onslaughts, price to the industry can be an issue of dispute.

While the basic logic of pricing of iron ore in India on cost-plus- reasonable gross margin basis for inter se competitiveness of Indian steel industry has rationale, considering the prevailing system and compatibility with international trade practice, this system can hardly be adopted. Therefore, all the iron and steel industries in India should have captive iron ore resources expeditiously corresponding to their capacity, to not only have level playing fields, but also sustain the growth rate of our steel industry for national economic sustainability.

Besides, export of iron ore may be restrained, in the national interest of conservation. All the integrated steel plants, capable of undertaking viable, scientific and efficient iron ore mining, should have access to captive iron ore mines. Give your observations on the case study.

*(**Source:** Based on an article that appeared in the Hindu Business Line, January 3, 2012)*

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